

Lab1 Jooho Kim

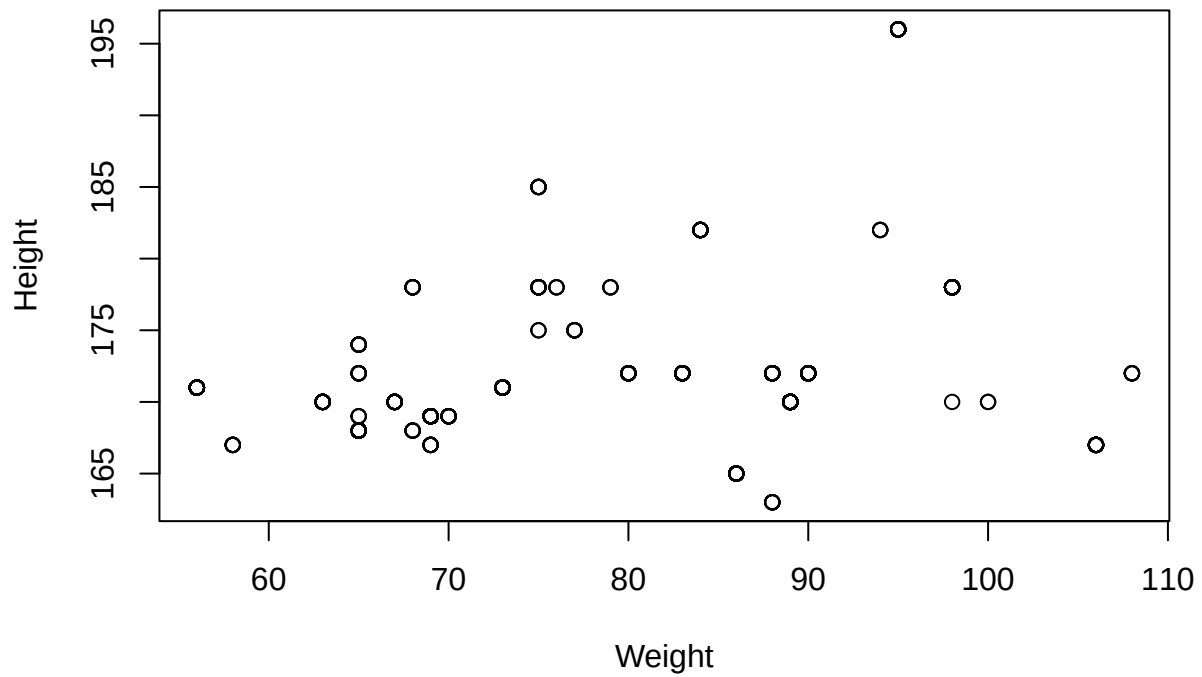
2025-09-21

```
df = read.csv("Absenteeism_at_work.csv", sep=";", header=TRUE)
head(df)
```

```
##   ID Reason.for.absence Month.of.absence Day.of.the.week Seasons
## 1 11                26                7                3        1
## 2 36                 0                7                3        1
## 3  3                23                7                4        1
## 4  7                 7                7                5        1
## 5 11                23                7                5        1
## 6  3                23                7                6        1
##   Transportation.expense Distance.from.Residence.to.Work Service.time Age
## 1                    289                    36            13    33
## 2                    118                    13            18    50
## 3                    179                    51            18    38
## 4                    279                     5            14    39
## 5                    289                    36            13    33
## 6                    179                    51            18    38
##   Work.load.Average.day Hit.target Disciplinary.failure Education Child
## 1                    239.554          97                0          1     2
## 2                    239.554          97                1          1     1
## 3                    239.554          97                0          1     0
## 4                    239.554          97                0          1     2
## 5                    239.554          97                0          1     2
## 6                    239.554          97                0          1     0
##   Social.drinker Social.smoker Pet Weight Height Body.mass.index
## 1              1             0   1   90   172             30
## 2              1             0   0   98   178             31
## 3              1             0   0   89   170             31
## 4              1             1   0   68   168             24
## 5              1             0   1   90   172             30
## 6              1             0   0   89   170             31
##   Absenteeism.time.in.hours
## 1                          4
## 2                          0
## 3                          2
## 4                          4
## 5                          2
## 6                          2
```

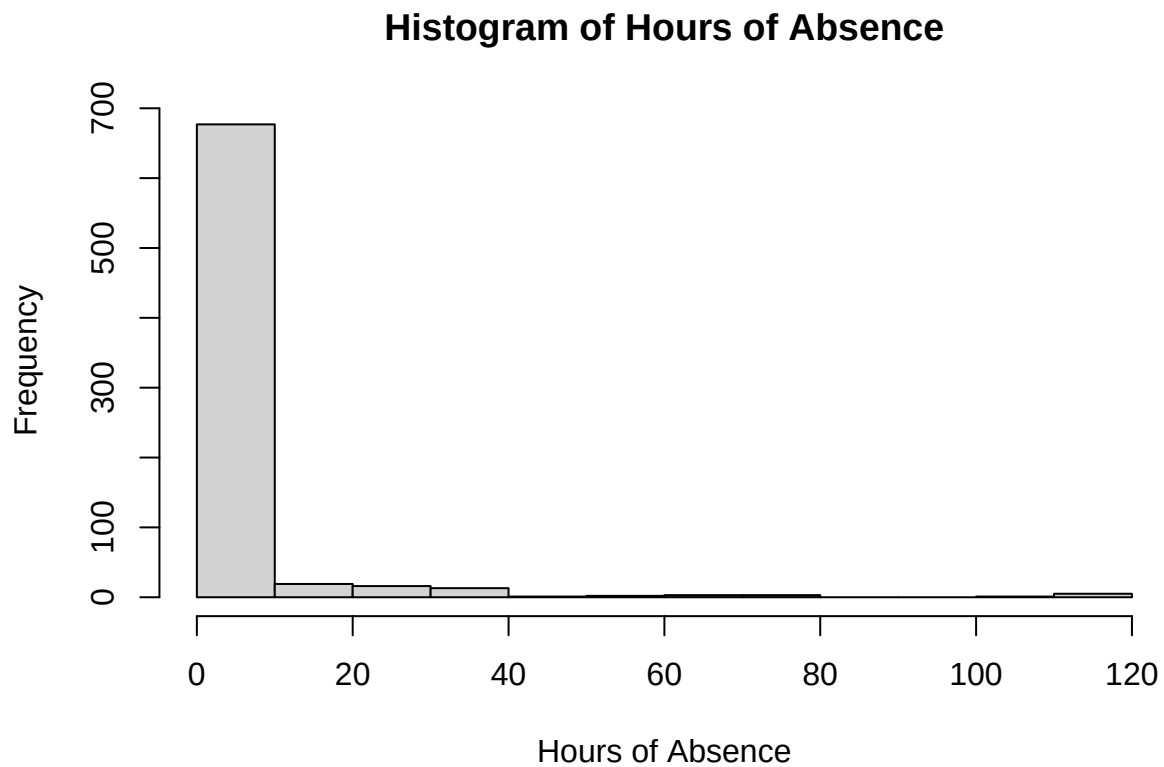
```
plot(df$Weight, df$Height,
     xlab = "Weight",
     ylab = "Height",
     main = "Scatter Plot of Height vs Weight")
```

Scatter Plot of Height vs Weight



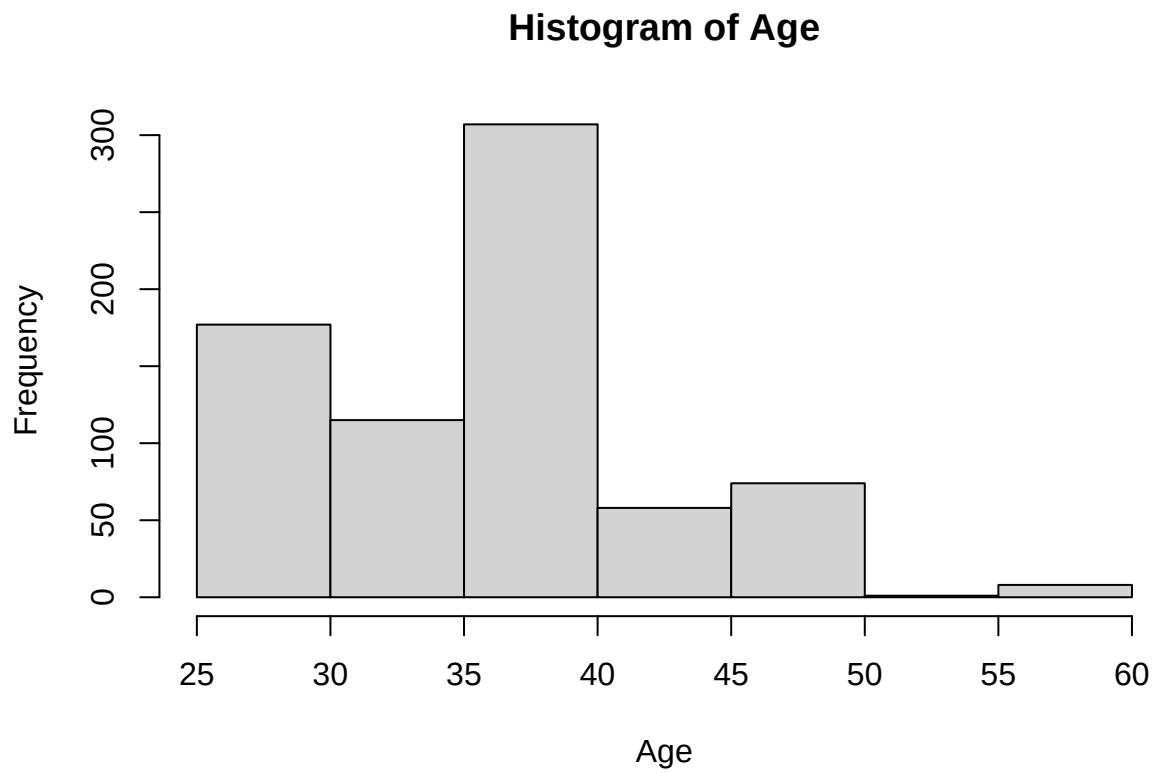
There does not appear to be a strong linear relationship between height and weight.

```
hist(df$Absenteeism.time.in.hours,  
      xlab = "Hours of Absence",  
      main = "Histogram of Hours of Absence")
```



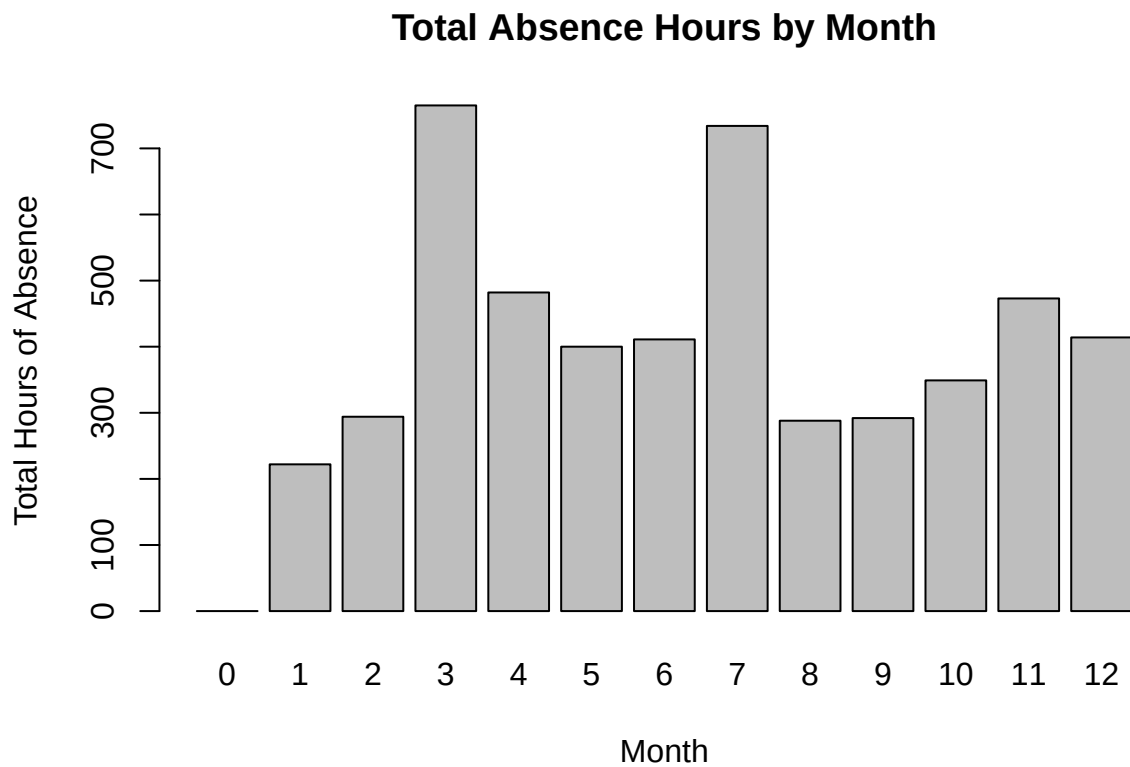
The histogram shows the distribution of absence hours, with most values clustered at lower hours.

```
hist(df$Age,  
      xlab = "Age",  
      main = "Histogram of Age")
```



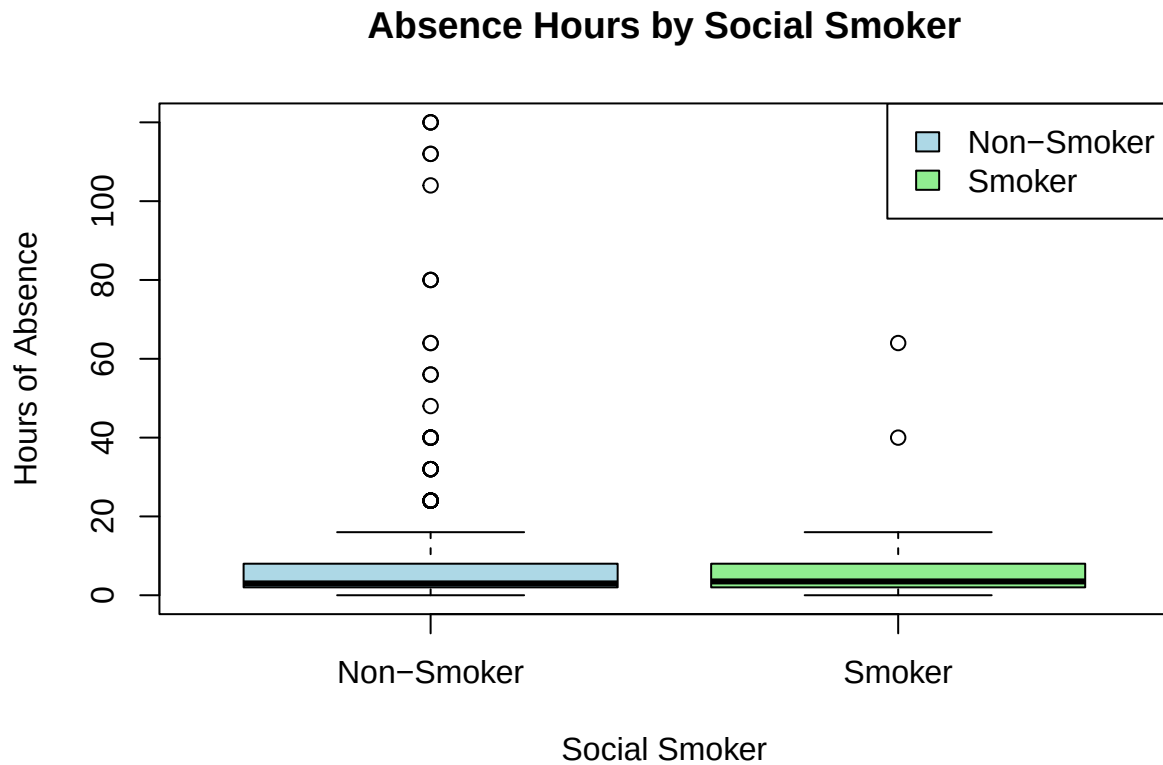
The histogram shows the age distribution of employees with recorded absences.

```
total_hours_by_month <- tapply(df$Absenteeism.time.in.hours,  
                               df$Month.of.absence,  
                               sum)  
  
barplot(total_hours_by_month,  
        xlab = "Month",  
        ylab = "Total Hours of Absence",  
        main = "Total Absence Hours by Month")
```



The bar plot shows how total absence hours vary across months.

```
boxplot(df$Absenteeism.time.in.hours ~ df$Social.smoker,  
        xlab = "Social Smoker",  
        ylab = "Hours of Absence",  
        main = "Absence Hours by Social Smoker",  
        col = c("lightblue", "lightgreen"),  
        names = c("Non-Smoker", "Smoker"))  
  
legend("topright",  
       legend = c("Non-Smoker", "Smoker"),  
       fill = c("lightblue", "lightgreen"))
```

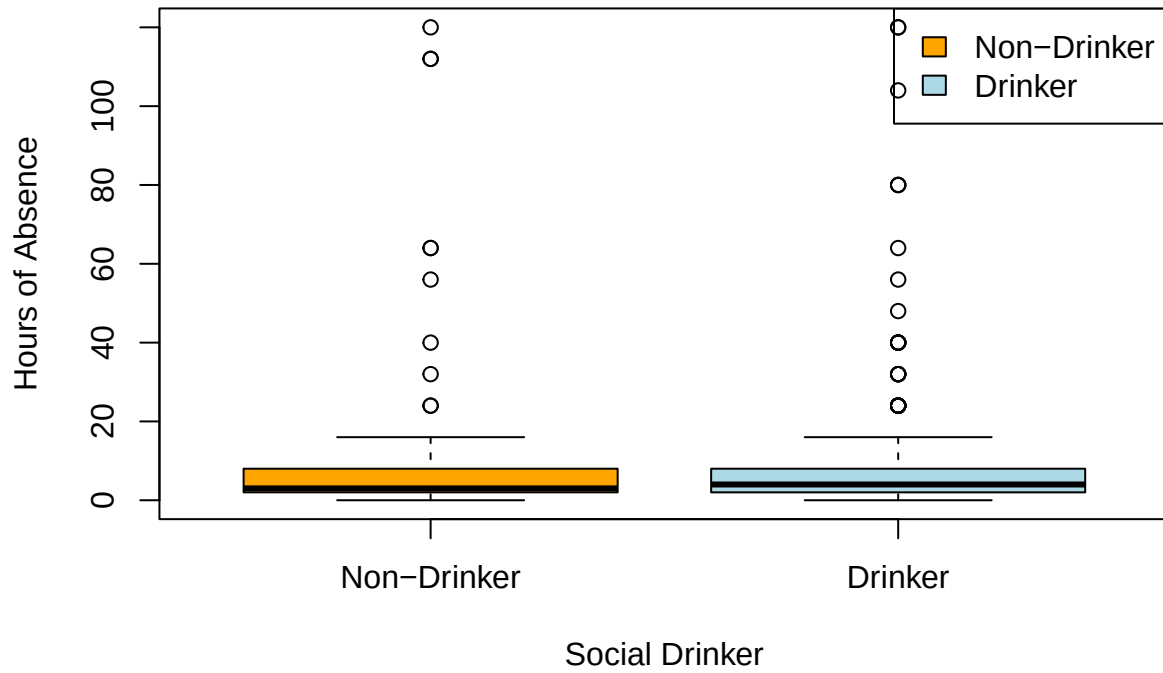


The box plot compares absence hours between social smokers and non-smokers.

```
boxplot(df$Absenteeism.time.in.hours ~ df$Social.drinker,
        xlab = "Social Drinker",
        ylab = "Hours of Absence",
        main = "Absence Hours by Social Drinker",
        col = c("orange", "lightblue"),
        names = c("Non-Drinker", "Drinker"))

legend("topright",
       legend = c("Non-Drinker", "Drinker"),
       fill = c("orange", "lightblue"))
```

Absence Hours by Social Drinker



The box plot compares absence hours between social drinkers and non-drinkers.